

Zachary Boroda

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EDUCATION

Brandeis University | Waltham, MA

Graduated May 2023

Major: Computer Science and Applied Mathematics, Phi Beta Kappa, Summa Cum Laude

GPA: 3.99/4.0

RESEARCH & PROFESSIONAL EXPERIENCE

Amazon Web Services (AWS) | Software Engineer II (promoted 2025) | Arlington, VA

August 2023 - Present

- Engineered high-throughput distributed system that marshals deletion processes across AWS service ecosystem affecting millions of AWS customers and achieving 99.99% availability for account lifecycle management
- Served as on-call engineer for a Tier 1 service, enhancing operational documentation and monitoring dashboards to improve incident response time
- Mentored two summer interns and one new hire, providing technical guidance and career development support

Kondev Group | Research Assistant | Waltham, MA

September 2021 - May 2025

- Investigating the structural differences between evolved logic gate models and logic gates generated through random walk processes under the supervision of Professor Jane Kondev
- Designed and implemented a theoretical evolution model in Python and deployed large-scale simulations on SLURM high-performance computing cluster
- Presented results to collaborators at HMS, BU and worked with a collaborator at Dartmouth
- Currently preparing paper for publication documenting methods and results

Brandeis University | Teaching Assistant | Waltham, MA

August 2020 - May 2023

- COSI 21A (Data Structures) with Professors Antonella Di Lillo and Iraklis Tsekourakis
- COSI 130A (Theory of Computation) with Professor James Storer
- Conducted biweekly office hours and led recitations for 100+ students, clarifying complex theoretical concepts and algorithms
- Graded assignments and exams, receiving high student feedback scores for clarity and accessibility of instruction

Johns Hopkins Applied Physics Lab | Internship | Laurel, MD

May 2021 - August 2021

- Integrated applied mathematics and physics with software development to solve aerospace engineering problems
- Built an optimized API for coordinate transformations between reference frames, improving computational efficiency for flight trajectory analysis systems

QBRc Lab | Research Rotations | Waltham, MA

May 2021 - August 2021

- Studied Drosophila flies and the effect of a mutation on their circadian rhythms
- Modeled complex systems studying their underlying dynamics with the SINDy Algorithm

AWARDS

Michtom Prize for Academic Excellence in Computer Science

May 2023

Awarded for exceptional creativity, innovation, and contributions benefiting the CS department and future students

EXTRACURRICULARS

Brandeis Mountaineering and Outdoors Club | Treasurer

August 2019 - May 2023

- Managed fundraising, budget allocation, and procurement for 200-member outdoor recreation club

BotNy — Plant analysis tool: Winner of Codestellation Hackathon 2020

2020

- Developed full-stack web application integrating Arduino sensor data with web-scraped plant care guidelines to provide real-time environmental monitoring and care recommendations